

The nonmetrizable finite-index expectation question

Literature-already-answered note for arXiv:2409.03531

11 August 2026

Status. Answered negatively in the direct follow-up arXiv:2409.17807. The source paper proves existence over compact metrizable bases; the follow-up constructs a compact nonmetrizable base and a continuous unital subhomogeneous commutative bundle for which there is no conditional expectation at all. Thus the unrestricted compact-Hausdorff question has a negative answer, while the metrizable theorem remains sharp in a meaningful selection-theoretic sense.

The question and the residue left by the source paper

Blanchard–Gogić Problem 3.11, restated in [1], asks whether a continuous unital subhomogeneous C^* -bundle $\mathcal{A} \twoheadrightarrow X$ over compact Hausdorff X always makes the central inclusion

$$C(X) \hookrightarrow \Gamma(\mathcal{A})$$

admit a finite-index conditional expectation. Here finite index means that for an expectation E there is a finite K with $KE - \text{id} \geq 0$. It also asks whether one can attain the optimal constant $K(E) = r(\mathcal{A})$.

The source paper answers the existence part affirmatively when X is compact metrizable (Theorem 2.12), and gives rank-3 counterexamples to the optimal constant. Its separate homogeneous-Banach-bundle question is also answered affirmatively and quantitatively there. The genuine residue was therefore: does an expectation always exist when the compact Hausdorff base is allowed to be nonmetrizable?

The later counterexample theorem

For a compact Hausdorff space Z and $n \geq 1$, let

$$Z_{[n]} = \{S \subseteq Z : 1 \leq |S| \leq n\}$$

with its Vietoris topology, and form the incidence space

$$Z_{[n]}^{\subseteq} = \{(z, S) \in Z \times Z_{[n]} : z \in S\}.$$

Projection onto S is an open continuous surjection $\pi : Z_{[n]}^{\subseteq} \twoheadrightarrow Z_{[n]}$ with fibers of size at most n , hence a classical branched cover.

Theorem 1.3 of [2] states that if $n \geq 3$ and Z contains the one-point compactification Δ^+ of an uncountable discrete space Δ , then the dual unital inclusion

$$C(Z_{[n]}) \xrightarrow{\pi^*} C(Z_{[n]}^{\subseteq}) \tag{1}$$

admits *no conditional expectation*. In bundle language, the fiber at S is the finite-dimensional abelian algebra $C(S) \cong \mathbb{C}^{|S|}$, so (1) is precisely a continuous unital subhomogeneous $C(Z_{[n]})$ -algebra. It is therefore a counterexample in the exact generality of Problem 3.11(a), stronger than needed because even an infinite-index expectation is absent.

Idea of the counterexample

An expectation $E : C(Z_{[3]}^{\subseteq}) \rightarrow C(Z_{[3]})$ determines, at each finite set $S \in Z_{[3]}$, a state on the fiber $C(S)$. Identifying points of Z with their evaluation characters in the weak-* dual of $C(Z)$, this is a probability measure

$$\mu_S \in \text{conv}(S).$$

Continuity of E forces $S \mapsto \mu_S$ to be a continuous selection of the lower-semicontinuous multifunction $S \mapsto \text{conv}(S)$.

For $Z = \Delta^+ = \Delta \sqcup \{\infty\}$ with Δ uncountable discrete, a classical Michael-type selection obstruction rules out such a selection on the triples $\{\delta, \delta', \infty\}$. In concrete terms, continuity near each pair involving ∞ forces certain sets of permissible partners to be cofinite. An uncountable pigeonhole argument produces one point lying in infinitely many of those cofinite sets; applying the same continuity constraint in the reverse direction forces its own set to omit infinitely many points, a contradiction. Restricting a hypothetical selection reduces the cases $Z \supseteq \Delta^+$ and $n \geq 3$ to this $Z = \Delta^+$, $n = 3$ case. Consequently no expectation can exist.

Exact scope after the two papers

Issue	Status
Existence over compact metrizable X	Affirmative in [1, Theorem 2.12].
Existence over arbitrary compact Hausdorff X	Negative by the incidence example in [2, Theorem 1.3].
Optimal equality $K(E) = r(\mathcal{A})$	Negative in general already in [1]; counterexamples begin at rank 3.
Homogeneous Banach-bundle unitarizability	Affirmative, quantitatively, in [1].
Extremally disconnected compact bases	Positive with optimal $K = r(\mathcal{A})$ in [2, Proposition 1.16].

The positive extremally disconnected result is worth separating from the negative answer: projectivity of such a compact space splits the surjection from the compact fiber-state space onto the base, yielding a continuous state choice and hence an expectation. The answer is thus not simply “nonmetrizable bases fail”; the obstruction is tied to continuous selection, and substantial nonmetrizable positive classes remain.

Novelty status. This packet records an exact literature resolution, not a new theorem from the run. During the upgrade search, the extremally disconnected argument was independently recovered, but Proposition 1.16 of the same follow-up already contains it, so no new partial result is claimed.

Human review. Check the restated question and metrizable theorem in source PDF pages 2–3; check the counterexample statement and selection reduction on supporting PDF page 5, the explicit Problem 3.11 remark on page 6, and the extremally disconnected result on pages 15–16. The conclusion depends on the continuity of the bundle and nonzero finite-dimensional fibers, all of which hold for the incidence construction.

References

- [1] A. Chirvasitu, *Non-commutative branched covers and bundle unitarizability*, arXiv:2409.03531. See the restatement of Problem 3.11 and Theorem 2.12.
- [2] A. Chirvasitu, *Finite-index phenomena and the topology of bundle singularities*, arXiv:2409.17807. See Theorem 1.3, the remarks immediately following it, and Proposition 1.16.